```
Feature names:
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']

Target names:
['setosa' 'versicolor' 'virginica']

Shape of binary dataset:
(100, 4)

Classes used:
['setosa' 'versicolor']

Accuracy:
1.0

Confusion Matrix:
[[10  0]
 [ 0 10]]

Classification Report:
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        10
  Versicolor       1.00      1.00      1.00        10

    accuracy                           1.00        20
   macro avg       1.00      1.00      1.00        20
weighted avg       1.00      1.00      1.00        20

code by ROHIT AHIR
```

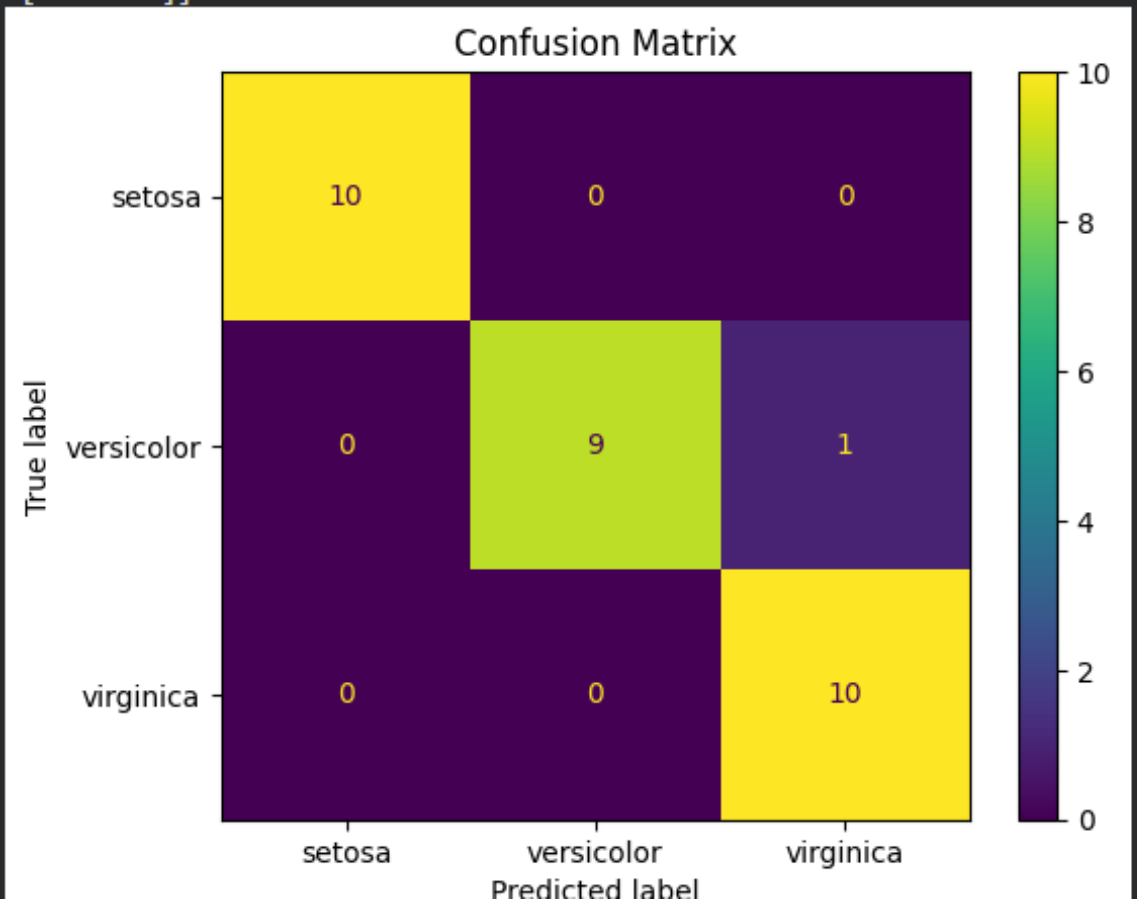
```
Feature Names:
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']

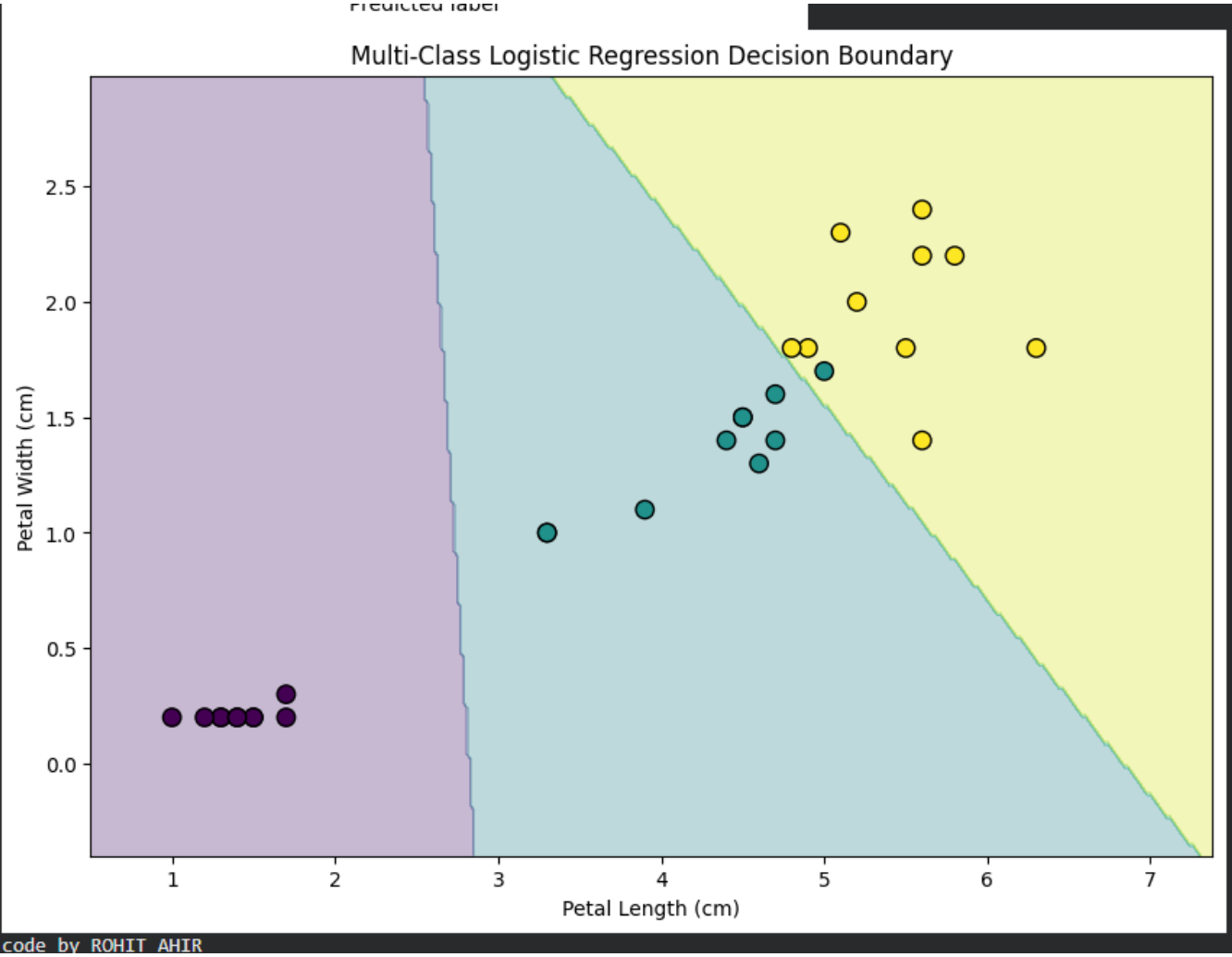
Target Names:
['setosa' 'versicolor' 'virginica']

Selected Features:
Petal Length
Petal Width

Accuracy:
0.9666666666666667

Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  0 10]]
```





code by ROHIT AHIR

```
Addition:
[[ 6  8]
 [10 12]]

Subtraction:
[[-4 -4]
 [-4 -4]]

Matrix Multiplication:
[[19 22]
 [43 50]]

Element-wise Multiplication:
[[ 5 12]
 [21 32]]

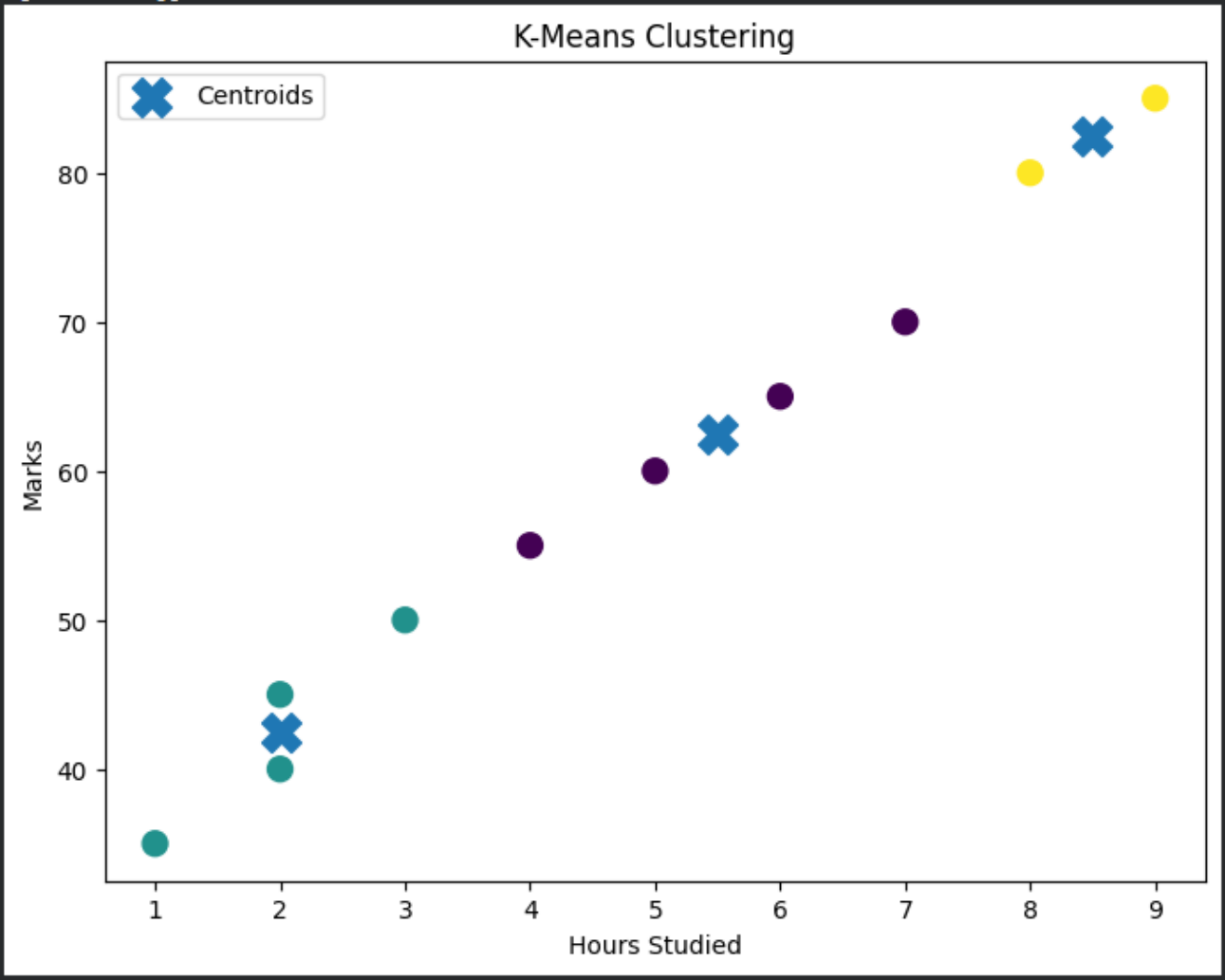
Transpose of A:
[[1 3]
 [2 4]]

Determinant of A:
-2.0000000000000004

Inverse of A:
[[-2.   1. ]
 [ 1.5 -0.5]]
code by ROHIT AHIR
```

```
Cluster Labels:
[1 1 1 1 0 0 0 0 2 2]

Centroids:
[[ 5.5 62.5]
 [ 2.  42.5]
 [ 8.5 82.5]]
```



code by ROHIT AHIR

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
/usr/local/lib/python3.13/dist-packages/keras/src/layers/reshaping/flatten.py:37: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first l
  super().__init__(**kwargs)
Model: "sequential"
```

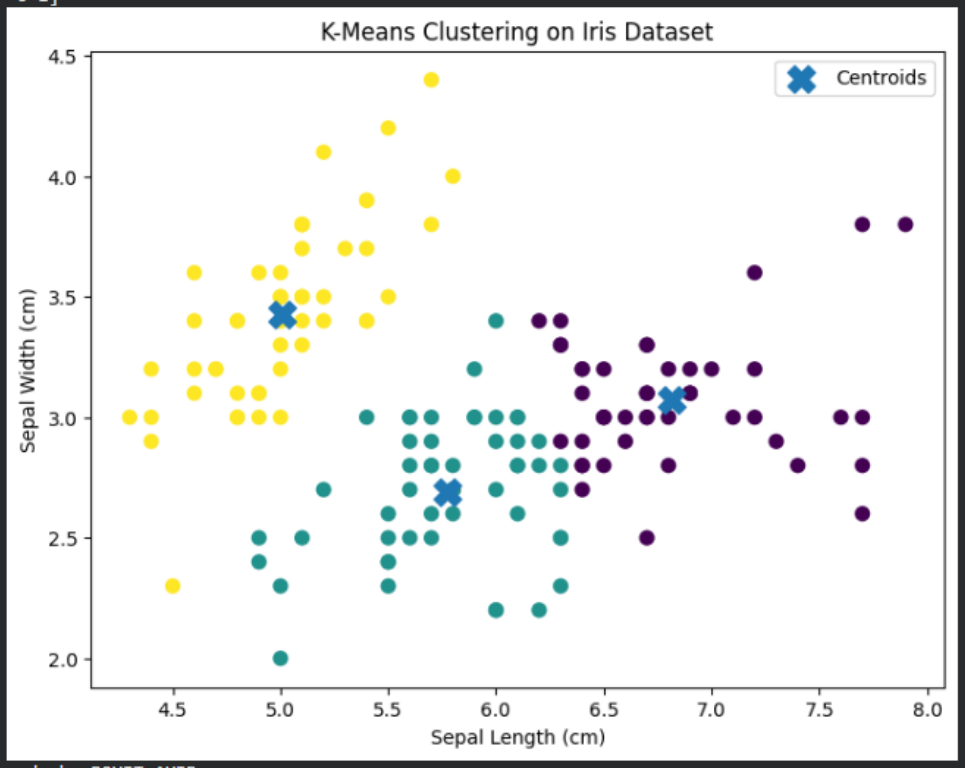
Layer (type)	Output Shape	Param #
flatten (Flatten)	(None, 784)	0
dense (Dense)	(None, 128)	100,480
dense_1 (Dense)	(None, 10)	1,290

```
 Total params: 101,770 (397.54 KB)
 Trainable params: 101,770 (397.54 KB)
 Non-trainable params: 0 (0.00 B)
Epoch 1/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 8s 5ms/step - accuracy: 0.9202 - loss: 0.2832 - val_accuracy: 0.9600 - val_loss: 0.1446
Epoch 2/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 6s 4ms/step - accuracy: 0.9644 - loss: 0.1234 - val_accuracy: 0.9667 - val_loss: 0.1138
Epoch 3/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 7s 5ms/step - accuracy: 0.9757 - loss: 0.0830 - val_accuracy: 0.9711 - val_loss: 0.0960
Epoch 4/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 11s 5ms/step - accuracy: 0.9808 - loss: 0.0619 - val_accuracy: 0.9730 - val_loss: 0.0888
Epoch 5/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 6s 4ms/step - accuracy: 0.9862 - loss: 0.0464 - val_accuracy: 0.9745 - val_loss: 0.0866
Epoch 6/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 8s 5ms/step - accuracy: 0.9897 - loss: 0.0366 - val_accuracy: 0.9744 - val_loss: 0.0843
Epoch 7/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 6s 4ms/step - accuracy: 0.9909 - loss: 0.0300 - val_accuracy: 0.9734 - val_loss: 0.0940
Epoch 8/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 7s 5ms/step - accuracy: 0.9935 - loss: 0.0239 - val_accuracy: 0.9735 - val_loss: 0.0986
Epoch 9/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 6s 4ms/step - accuracy: 0.9948 - loss: 0.0183 - val_accuracy: 0.9751 - val_loss: 0.0946
Epoch 10/10
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 9s 6ms/step - accuracy: 0.9957 - loss: 0.0149 - val_accuracy: 0.9765 - val_loss: 0.0885
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9774 - loss: 0.0792
Test Accuracy: 0.977400004863739
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step
Predicted digit: 7
Actual digit: 7
```

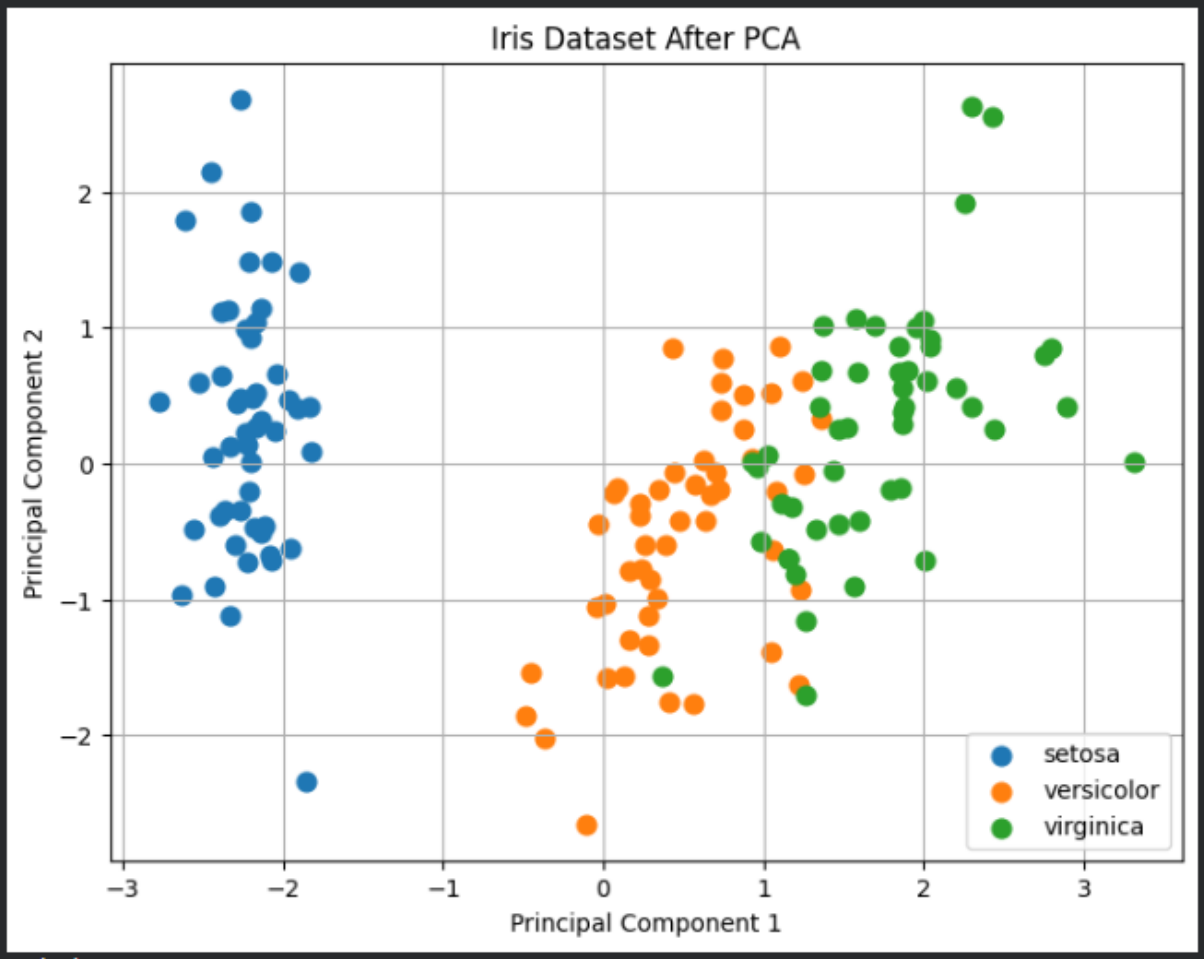


```
Cluster Centers:
[[6.81276596 3.07446809]
 [5.77358491 2.69245283]
 [5.006      3.428     ]]

Cluster Labels:
[2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
 2 2 2 2 2 2 2 2 2 2 2 2 2 2 0 0 0 1 0 1 0 1 0 1 1 1 1 1 1 0 1 1 1 1 1 1 1
 0 0 0 0 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 0 1 0 0 0 0 1 0 0 0 0
 0 0 1 1 0 0 0 0 1 0 1 0 1 0 0 1 1 0 0 0 0 0 1 1 0 0 0 1 0 0 0 1 0 0 0 1 0
 0 1]
```

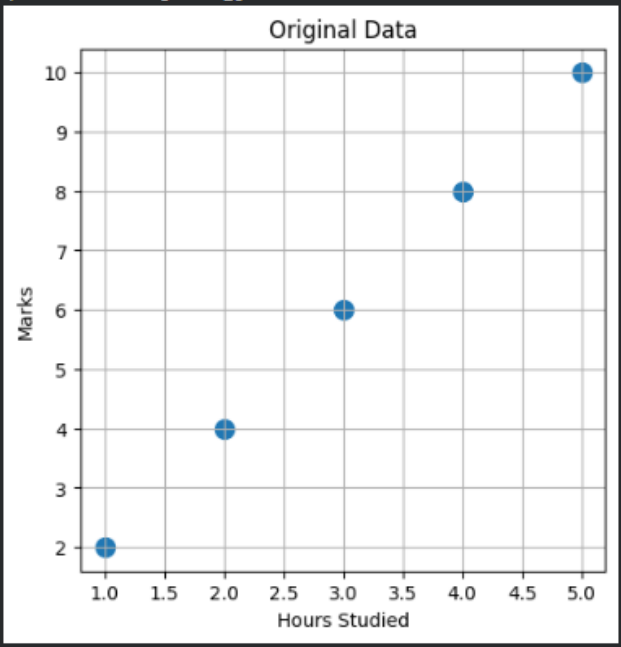


code by ROHIT AHIR



code by ROHIT AHIR

```
Original Data:
   Hours_Studied  Marks
0              1      2
1              2      4
2              3      6
3              4      8
4              5     10
```




```
Mean:
[3. 6.]

Centered Data:
[[-2. -4.]
 [-1. -2.]
 [ 0.  0.]
 [ 1.  2.]
 [ 2.  4.]]

Covariance Matrix:
[[ 2.5  5. ]
 [ 5.  10. ]]

Eigenvalues:
[ 0.  12.5]

Eigenvectors:
```

```
Eigenvalues:
[ 0.  12.5]

Eigenvectors:
[[-0.89442719 -0.4472136 ]
 [ 0.4472136  -0.89442719]]

Sorted Eigenvalues:
[12.5  0. ]

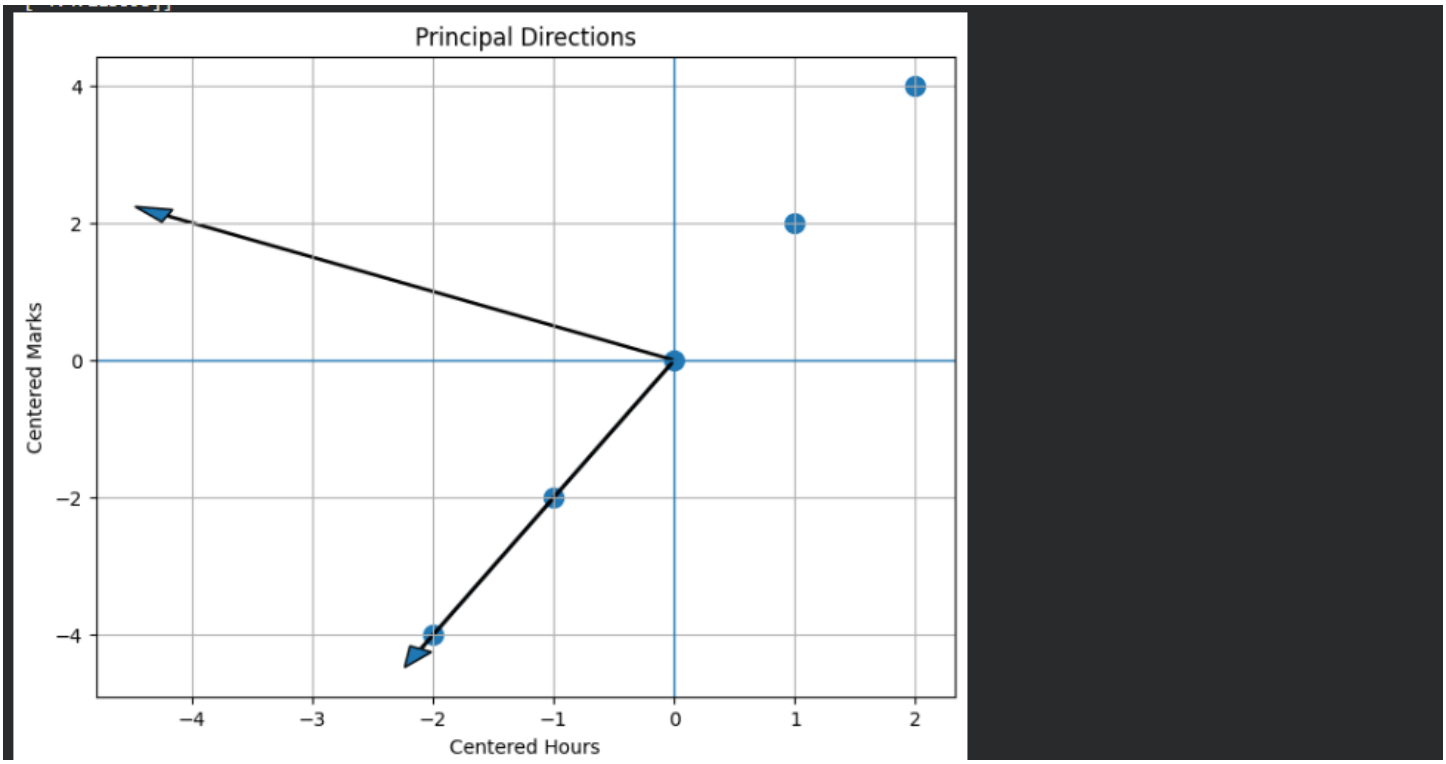
Sorted Eigenvectors:
[[-0.4472136  -0.89442719]
 [-0.89442719  0.4472136 ]]

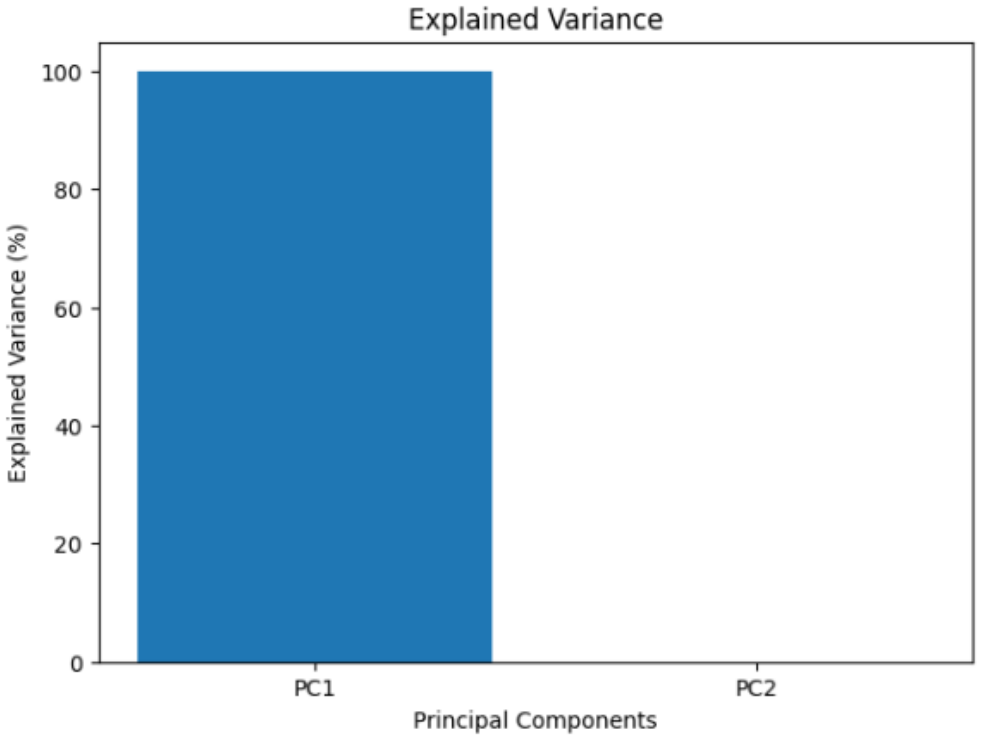
Explained Variance Ratio:
[1. 0.]

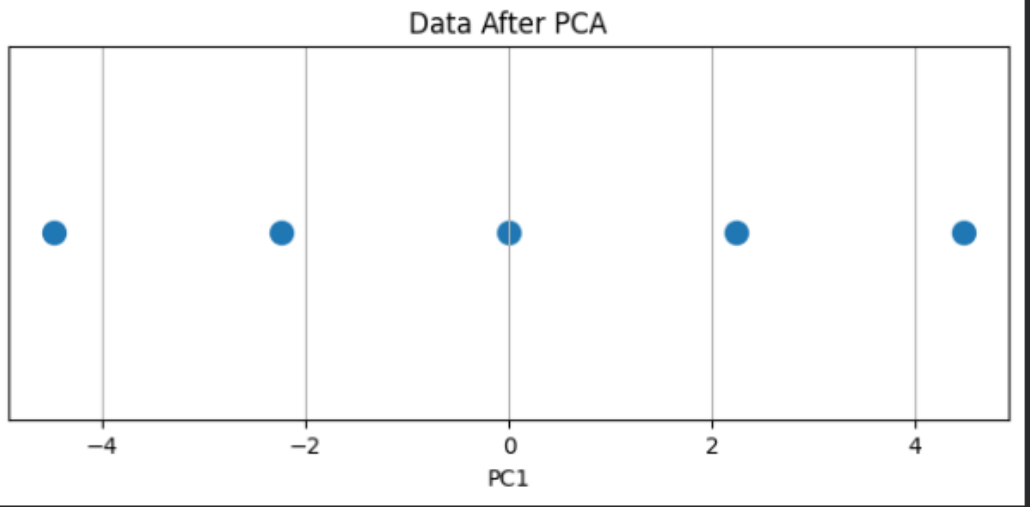
Explained Variance Percentage:
[100.   0.]

Selected Components:
[[-0.4472136 ]
 [-0.89442719]]

PCA Transformed Data:
[[ 4.47213595]
 [ 2.23606798]
 [ 0.        ]
 [-2.23606798]
 [-4.47213595]]
```







code by ROHIT AHIR